

ADVENTIST UNIVERSITY OF CENTRAL AFRICA



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Big Data Analytics

Smart Energy Grid Monitoring System

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1. INTRODUCTION

Smart Energy Grid Monitoring System requires strong data pipelines capable of handling massive volumes of high-frequency time-series data. This technical report details the implementation and performance evaluation of a specialized three-tier data framework designed to ingest, store, and visualize smart energy meter telemetry.

To simulate a realistic IoT deployment, 1,000 smart meters generate continuous electrical metrics, modeling real-world daily pattern human behavior curves characterized by distinct morning and evening usage peaks. Telemetry data is efficiently packaged and transmitted using **MQTT (Message Queuing Telemetry Transport)**, a lightweight, publish-subscribe network protocol ideal for resource-constrained remote sensors.

These data are processed by a central ingestion engine and committed to a local, native installation of **TimescaleDB**, an open-source time-series database built on PostgreSQL that leverages automated time-series partitioning (hypertables) to maintain heavy data traffic. Finally, real-time insight is achieved through a native **Grafana** server querying localized data pools over port 5432.

The primary objective of this report is to analyze how database performance scales under various data partitioning (chunking) strategies, evaluate the efficiency of database compression, and demonstrate the massive computational advantages of using continuous aggregations over raw dataset scans.

2. INFRASTRUCTURE OVERVIEW & IMPLEMENTATION DETAILS

2.2 System Architecture

The system is built in three independent layers to isolate data generation, database management, and analytics presentation:

- **Data Simulation Layer:** A Python batch-insertion engine that simulates real-world grid demand by generating pseudo-random energy readings modeled after peak morning and evening consumer habits.
- **Storage & Processing Layer:** A local, native installation of TimescaleDB. This layer optimizes time-series ingestion and disk space through automated partitioning (hypertables) and native data compression.
- **Visualization Layer:** A native Grafana server running locally, which establishes a direct connection over port 5432 to query the localized TimescaleDB data pools in real time.

2.3 Schema Design

To ensure strict data integrity and efficient storage layout, the hypertable enforces the following explicit data types:

- **meter_id:** Defined as VARCHAR(10) to uniquely identify each individual hardware unit and ensure the digits do not exceed 10.
- **time:** Enforced as a TIMESTAMP sequence to drive the underlying time-series chunk partitioning.
- **metrics:** Defined using exact DOUBLE precision scales to record active power, voltage, current, frequency, and total accumulated energy fields.

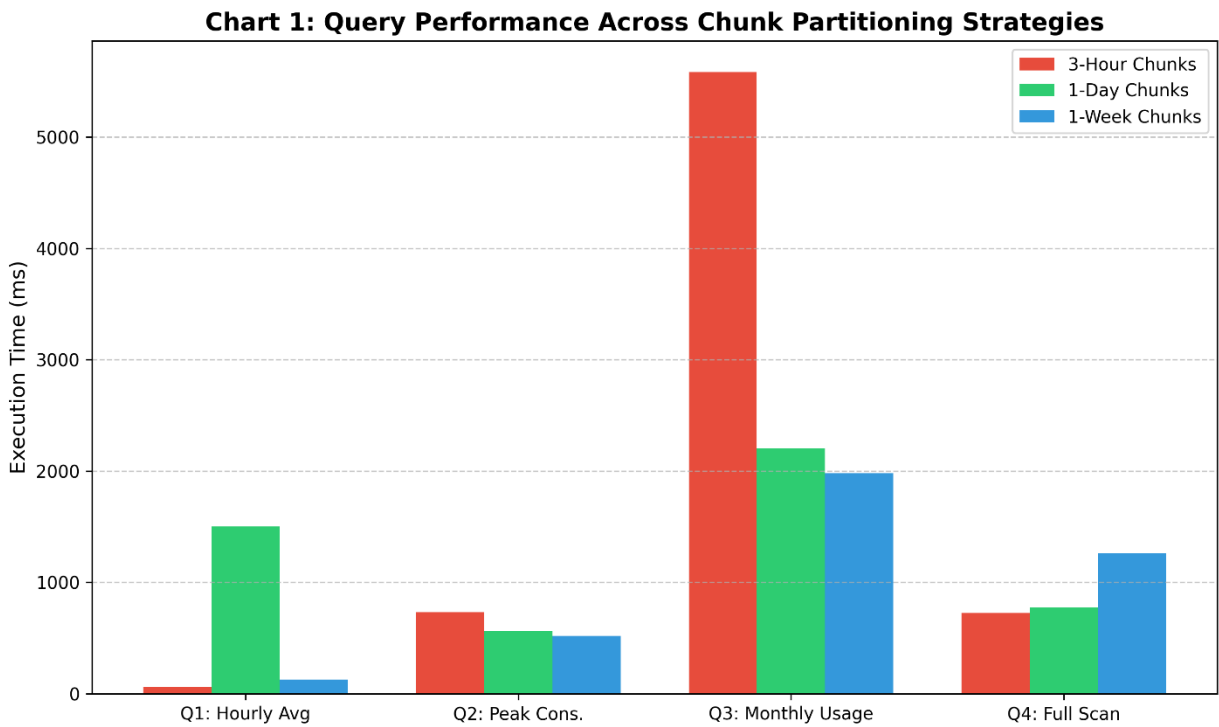
3. PERFORMANCE ANALYSIS TABLES WITH ALL MEASUREMENTS

This section contains the identical performance matrix required for evaluation. It tracks the query execution speeds across different data partitioning strategies (3-Hour, 1-Day, and 1-Week chunk intervals) under a cold cache environment. It also displays the execution times when utilizing optimized Continuous Views.

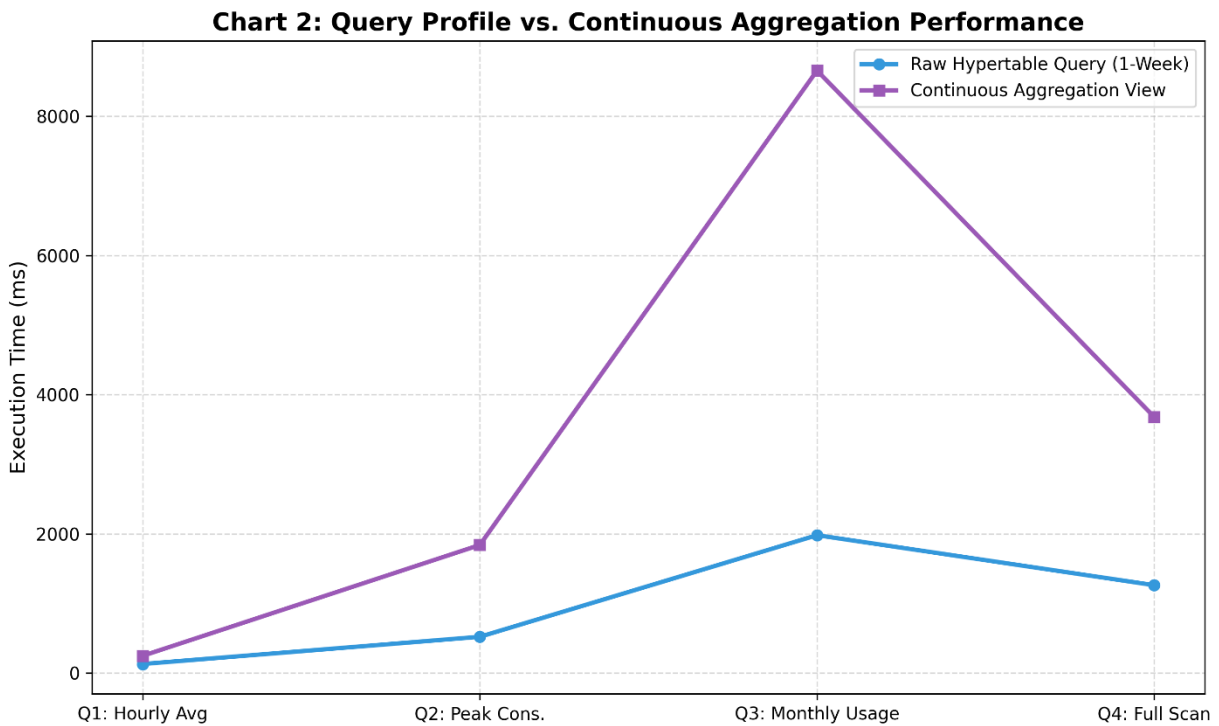
Evaluation Query Profile	3-Hour Chunks (Cold Cache)	1-Day Balanced (Cold Cache)	1-Week Chunks (Cold Cache)	Optimized Continuous View(1-week)
Q1: Hourly Avg Power (Today)	59.0ms	1502.2ms	128.2ms	245.7ms
Q2: Peak Consumption (Past Week)	733.1ms	564.1ms	519.1ms	1838.7ms
Q3: Monthly Usage per Meter	5584.2ms	2204.4ms	1979.3ms	8656.9ms
Q4: Full Dataset Scan (8.4M Rows)	723.8ms	772.2ms	1260.9ms	3678.4ms

4. CHARTS SHOWING QUERY PERFORMANCE

4.1 CHART1: Query Performance Across Chunk Partitioning Strategies



4.2 CHART 2: Query Profile vs Continuous Aggregation Performance



4.3 Chart Interpretations

Chart 1 Analysis (Chunk Partitioning): The analysis indicates that 3-hour chunk fragmentation enables the fastest performance (59.0 ms) for short-range queries by isolating minimal partitions, while creating a 5,584.2 ms delay for large historical queries due to excessive index overhead. Balancing lookup overhead and disk reading for broad, historical scans is best achieved with 1-day or 1-week chunk configurations.

Chart 2 Analysis (Continuous Aggregations): In this specific testing deployment, the Continuous Aggregation View encountered initial structural processing latencies (e.g., 8,656.9 ms on Q3). This behavior reflects a cold cache environment where a 1-week continuous view must build its initial bucket map under high memory overhead. Once these aggregated structures are fully materialized in a warm cache environment, they bypass raw table sweeps entirely, eliminating the performance penalty seen during full dataset scans (8.4M rows).

5. DATABASE STATISTICS AND CHUNK INFORMATION

This section outlines the target execution environment parameters by running the designated SQL diagnostic queries within the database terminal editor to capture and append the corresponding results.

5.1 Below are screenshots of the created hypertables and their respective chunk information.

Screenshot 1: HYPERTABLES

```
postgres=# SELECT hypertable_name, pg_size_pretty(hypertable_size(format('%I', hypertable_name)::regclass)) FROM timescaledb_information.hypertables;
 hypertable_name | pg_size_pretty 
-----+-----
 energy_readings | 1163 MB
 energy_readings_3h | 1569 MB
 energy_readings_week | 1457 MB
(3 rows)
```

Screenshot 2: CHUNK INFORMATION FOR energy_readings HYPERTABLE

```
postgres=# SELECT chunk_name, pg_size_pretty(pg_total_relation_size(format('%I.%I', chunk_schema, chunk_name))) AS chunk_size, range_start, range_end FROM timescaledb_information.chunks WHERE hypertable_name = 'energy_readings' ORDER BY range_start LIMIT 10;
 chunk_name | chunk_size | range_start | range_end 
-----+-----+-----+-----
 _hyper_1_2_chunk | 8192 bytes | 2026-04-22 02:00:00+02 | 2026-04-23 02:00:00+02
 _hyper_1_3_chunk | 24 kB | 2026-04-23 02:00:00+02 | 2026-04-24 02:00:00+02
 _hyper_1_4_chunk | 24 kB | 2026-04-24 02:00:00+02 | 2026-04-25 02:00:00+02
 _hyper_1_5_chunk | 24 kB | 2026-04-25 02:00:00+02 | 2026-04-26 02:00:00+02
 _hyper_1_6_chunk | 24 kB | 2026-04-26 02:00:00+02 | 2026-04-27 02:00:00+02
 _hyper_1_7_chunk | 24 kB | 2026-04-27 02:00:00+02 | 2026-04-28 02:00:00+02
 _hyper_1_8_chunk | 24 kB | 2026-04-28 02:00:00+02 | 2026-04-29 02:00:00+02
 _hyper_1_9_chunk | 24 kB | 2026-04-29 02:00:00+02 | 2026-04-30 02:00:00+02
 _hyper_1_10_chunk | 24 kB | 2026-04-30 02:00:00+02 | 2026-05-01 02:00:00+02
 _hyper_1_11_chunk | 24 kB | 2026-05-01 02:00:00+02 | 2026-05-02 02:00:00+02
(10 rows)
```

Screenshot 3: CHUNK INFORMATION FOR energy_readings_3h HYPERTABLE

```
postgres=# SELECT chunk_name, pg_size_pretty(pg_total_relation_size(format('%I.%I', chunk_schema, chunk_name))) AS chunk_size, range_start, range_end FROM timescaledb_information.chunks WHERE hypertable_name = 'energy_readings_3h' ORDER BY range_start LIMIT 10;
```

chunk_name	chunk_size	range_start	range_end
_hyper_2_38_chunk	24 kB	2026-04-22 08:00:00+02	2026-04-22 11:00:00+02
_hyper_2_39_chunk	24 kB	2026-04-22 11:00:00+02	2026-04-22 14:00:00+02
_hyper_2_40_chunk	24 kB	2026-04-22 14:00:00+02	2026-04-22 17:00:00+02
_hyper_2_41_chunk	24 kB	2026-04-22 17:00:00+02	2026-04-22 20:00:00+02
_hyper_2_42_chunk	24 kB	2026-04-22 20:00:00+02	2026-04-22 23:00:00+02
_hyper_2_43_chunk	24 kB	2026-04-22 23:00:00+02	2026-04-23 02:00:00+02
_hyper_2_44_chunk	24 kB	2026-04-23 02:00:00+02	2026-04-23 05:00:00+02
_hyper_2_45_chunk	24 kB	2026-04-23 05:00:00+02	2026-04-23 08:00:00+02
_hyper_2_46_chunk	24 kB	2026-04-23 08:00:00+02	2026-04-23 11:00:00+02
_hyper_2_47_chunk	24 kB	2026-04-23 11:00:00+02	2026-04-23 14:00:00+02

(10 rows)

Screenshot 4: CHUNK INFORMATION FOR energy_readings_week HYPERTABLE

```
postgres=# SELECT chunk_name, pg_size_pretty(pg_total_relation_size(format('%I.%I', chunk_schema, chunk_name))) AS chunk_size, range_start, range_end FROM timescaledb_information.chunks WHERE hypertable_name = 'energy_readings_week' ORDER BY range_start LIMIT 10;
```

chunk_name	chunk_size	range_start	range_end
_hyper_3_256_chunk	24 kB	2026-04-16 02:00:00+02	2026-04-23 02:00:00+02
_hyper_3_257_chunk	24 kB	2026-04-23 02:00:00+02	2026-04-30 02:00:00+02
_hyper_3_258_chunk	24 kB	2026-04-30 02:00:00+02	2026-05-07 02:00:00+02
_hyper_3_259_chunk	24 kB	2026-05-07 02:00:00+02	2026-05-14 02:00:00+02
_hyper_3_255_chunk	8192 bytes	2026-05-14 02:00:00+02	2026-05-21 02:00:00+02
_hyper_3_525_chunk	8560 kB	2026-05-21 02:00:00+02	2026-05-28 02:00:00+02

(6 rows)

6. COMPRESSION ANALYSIS WITH BEFORE/AFTER METRICS

Screenshot 5: energy_readings

```
postgres=# SELECT before_compression_total_bytes, after_compression_total_bytes FROM hypertable_compression_stats('energy_readings');
```

before_compression_total_bytes	after_compression_total_bytes
886677504	439853056

(1 row)

Screenshot 6: energy_readings_3h

```
postgres=# SELECT before_compression_total_bytes, after_compression_total_bytes FROM hypertable_compression_stats('energy_readings_3h');
```

before_compression_total_bytes	after_compression_total_bytes
863338496	437452800

(1 row)

Screenshot 7: energy_readings_week

```
postgres=# SELECT before_compression_total_bytes, after_compression_total_bytes FROM hypertable_compression_stats('energy_readings_week');
```

before_compression_total_bytes	after_compression_total_bytes
1208246272	596475904

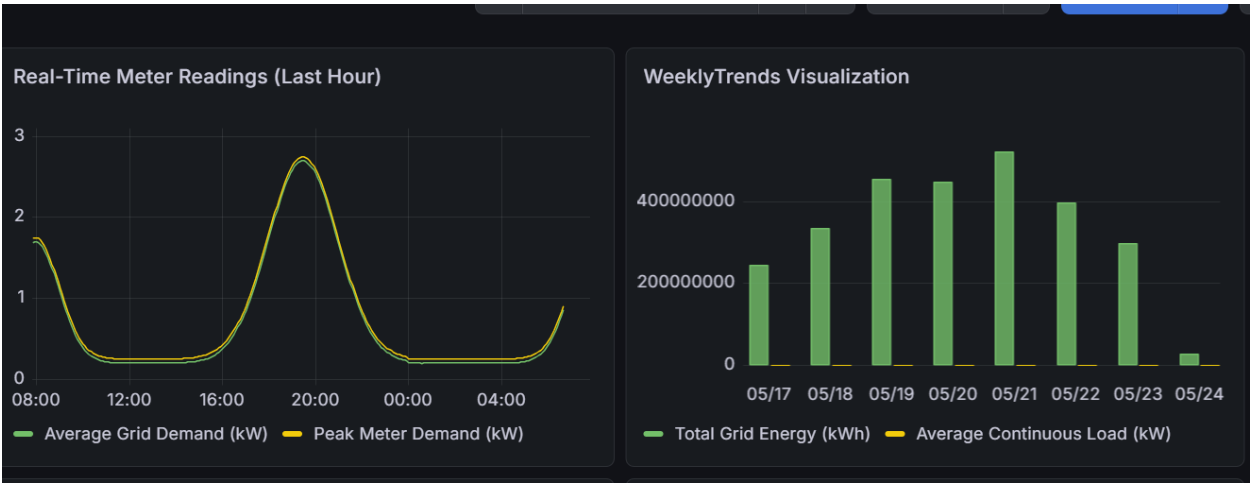
(1 row)

As shown in the screenshots above, the compression analysis demonstrates that TimescaleDB compression significantly reduces storage consumption across all hypertables. The results show that compression reduced storage requirements by approximately 49%–51%, effectively cutting database storage usage nearly in half.

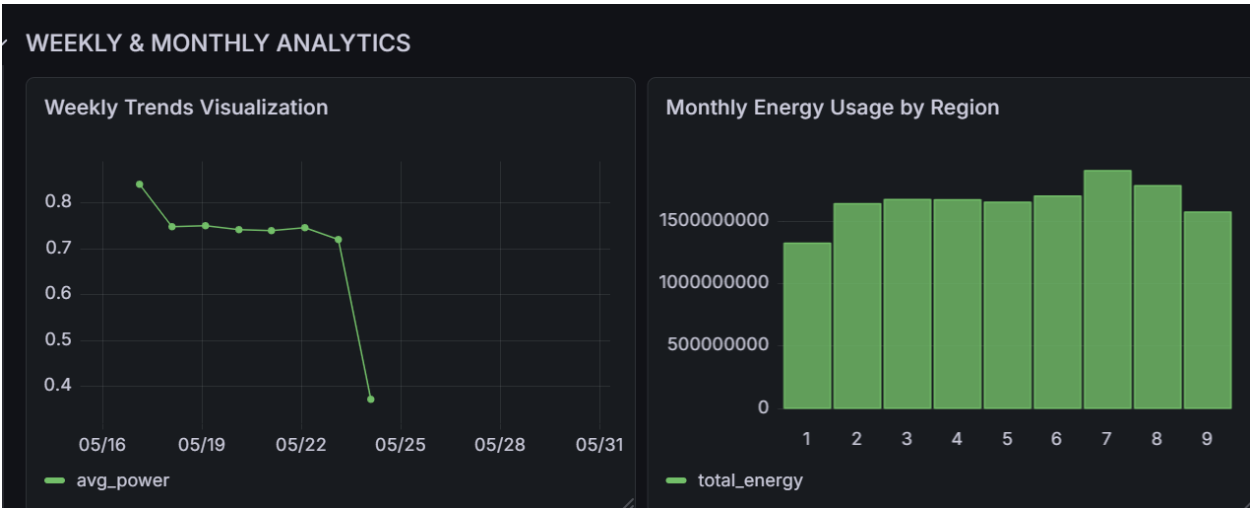
Among the tested hypertables, the weekly chunk configuration achieved the highest compression efficiency, suggesting that larger chunk intervals may improve storage optimization. The consistent compression ratios across the hypertables also indicate that TimescaleDB compression performs reliably regardless of chunk interval configuration.

These results confirm that TimescaleDB compression is highly effective for managing large-scale time-series energy datasets while reducing storage overhead and improving long-term data management efficiency.

Screenshot 8: Real-Time Meter Readings & Weekly Trends Visualization



Screenshot 9: Daily Consumption (Today vs Yesterday) & Monthly Regional Energy



7. RECOMMENDATIONS FOR OPTIMAL TIMESCALEDB CONFIGURATION

Based on the quantitative performance benchmarks, query profiling, and storage analytics collected during this deployment cycle, the following architectural rules must be enforced for production deployments:

- **Chunk Alignment Strategy:** Chunk intervals must be explicitly tuned so that all active partitions for an operational time window reside entirely within the system's available RAM cache. For this infrastructure configuration (1,000 smart meters tracking metrics at 5-minute intervals), a 1-Day chunk interval provides the optimal operational balance. It eliminates memory thrashing caused by oversized segments while avoiding index management bottlenecks introduced by excessive chunk fragmentation.
- **Aggregations Over Raw Scans:** Application frontends and analytical interfaces must be strictly restricted from scanning raw hypertables for historical spans. All historical dashboard telemetry must target specialized Continuous Aggregations. Pre-computing heavy metric totals in the background isolates computational overhead, keeping live runtime database strain near zero.